Mechanical Reasoning in Differential Geometry

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Abstract. We show how ideas from classical mechanics can be turned around to suggest and motivate results in differential geometry. Mechanical setups, when interpreted geometrically, lead naturally to statements such as: a shortest path on a surface is a geodesic, the Gauss lemma, Jacobi's theorem, and the isoperimetric inequality. While such arguments may feel informal to those who prefer formal rigor, they follow a tradition developed by Levi [1, 2], Tokieda [3, 4], Vallejo and Bove [5], Graham and Tokieda [6], and Kim [7, 8], whose works show that many find this kind of physically inspired reasoning intuitive and interesting, and it is in this spirit that the present work is offered.

Keywords:physical proof, differential geometry, geodesics, isoperimetric inequality

1. WORK AND POTENTIAL ENERGY OF A MOTOR SYSTEM Consider a motor that pulls a thread with constant tension T. Imagine a tiny worker inside the motor, holding the end of the thread and pulling it with constant force T (see Figure 1).

Outside the motor, the thread may pass through various pulleys, bend in multiple directions, or even rotate and oscillate. However, regardless of what happens outside the motor, the work done by the tiny worker on the segment of thread they are pulling depends only on the amount of displacement. If the worker pulls the thread by a distance Δs, while maintaining constant tension T, the work done is

$$W = T \cdot \Delta s.$$

Therefore, if the motor pulls the thread with constant tension T, and the total length of thread outside the motor changes by Δl, then the work done by the motor is

$$W = -T \cdot \Delta l.$$

It follows that the potential energy of the system can be written as

$$U = T\, l,$$

where l is the total length of the thread outside the motor.

2. GEODESICS

Minimum path is geodesic Throughout this article, the term "surface" refers to a 2-dimensional Riemannian manifold embedded in $\mathbb{R}^3$, and the term "curve" refers to a parametrized curve in $\mathbb{R}^3$, as we aim to illustrate theorems through physical reasoning.

Theorem 1. *Let S be a surface, and let $p, q \in S$ be two points. If a path of minimal length from p to q exists, then it is a geodesic.*

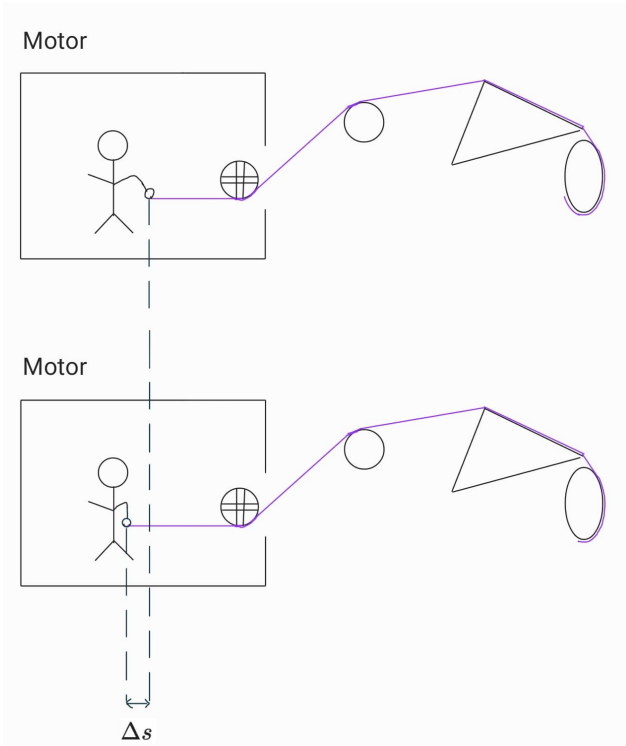

Figure 1. A motor pulls a thread with constant tension T. The internal "tiny worker" pulls the thread straight, while the external thread may bend through pulleys.

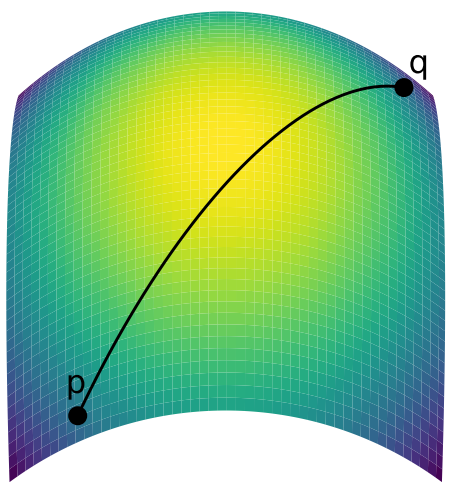

Figure 2. A minimal-length path is a geodesic.

While a rigorous proof of this fact is non-trivial (see [9, Section 5.4, Proposition 3]), classical mechanics provides a useful shortcut.

Proof. Consider a motor fixed at point p that pulls a thread (with nonzero mass) with a constant force of magnitude T, while the other end of the thread is fixed at point q.

The thread is constrained to lie on the surface, and the only forces acting on it are the constraint force and the tension. If we denote the length of the thread by l, then the potential energy of the system is

$$U = Tl, \tag{1}$$

as deduced in Section 1. Minimizing the length of the thread corresponds to minimizing the potential energy, so the system reaches equilibrium when the thread takes the minimal-length configuration.

Let the thread at equilibrium be described by a curve $\mathbf{r} : [0, L] \to S$ parametrized by arc length, such that $\mathbf{r}(0) = p$ and $\mathbf{r}(L) = q$. Let the tension at point $\mathbf{r}(s)$ be $T(s)$, and let $\mathbf{F}(s)$ denote the total constraint force applied to the portion $\mathbf{r}(u)$ for $0 \le u \le s$. If we define

$$\mathbf{f}(s) = \frac{d\mathbf{F}}{ds}(s),$$

i.e., $\mathbf{f}$ as the constraint force per unit length, then $\mathbf{f}(s)$ is normal to S because the thread element at $\mathbf{r}(s)$ can move on S without friction. We can thus write $\mathbf{f}(s) = f(s)\mathbf{N}(s)$, where $\mathbf{N}(s)$ is the unit normal vector at $\mathbf{r}(s)$.

Now consider the small portion of the thread $\mathbf{r}(u)$ for $s \le u \le s + \Delta s$. The net force on this segment is

$$-T(s)\mathbf{r}'(s) + T(s + \Delta s)\mathbf{r}'(s + \Delta s) + f(s)\mathbf{N}(s)\,\Delta s + o(\Delta s) = 0. \tag{2}$$

(See Figure 3.)

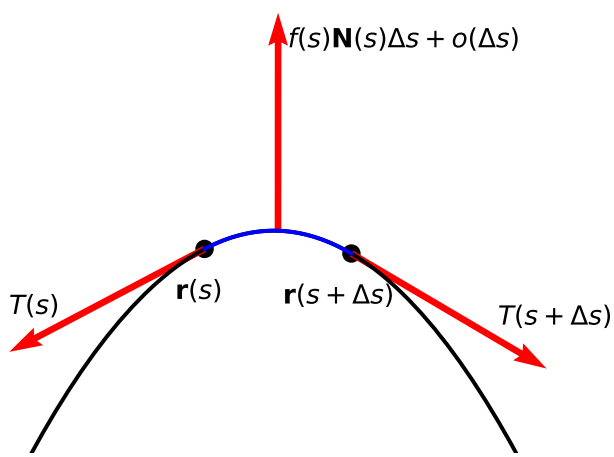

Figure 3. Force diagram for a small portion of the thread.

Dividing (2) by Δs and sending $\Delta s \to 0$ yields

$$T'(s)\mathbf{r}'(s) + T(s)\mathbf{r}''(s) + f(s)\mathbf{N}(s) = 0. \tag{3}$$

Taking the inner product of (3) with $\mathbf{r}'(s)$ gives $T'(s) = 0$, since $\mathbf{r}$ is parametrized by arc length, so $\mathbf{r}'(s) \cdot \mathbf{r}'(s) = 1$ and differentiating yields $\mathbf{r}'(s) \cdot \mathbf{r}''(s) = 0$. Thus $T(s)$ is constant: $T(s) \equiv T$. Substituting back gives

$$T\,\mathbf{r}''(s) + f(s)\mathbf{N}(s) = 0,$$

which implies that $\mathbf{r}''(s)$ is normal to the surface. Hence, the curve is a geodesic. ■

Gauss' Lemma

Theorem 2 (Gauss' Lemma). *Let S be a surface and $p \in S$. For any tangent vectors $\mathbf{x}, \mathbf{w} \in T_pS$ for which the terms in* (4) *are well-defined, the following identity holds:*

$$\mathbf{x} \cdot \mathbf{w} = (d\exp_p)_{\mathbf{x}}(\mathbf{x}) \cdot (d\exp_p)_{\mathbf{x}}(\mathbf{w}). \tag{4}$$

The rigorous and technical proof can be found in [9], Section 5.5, Lemma 2. Here, we present an intuitive mechanical proof.

Proof. Suppose a motor is fixed at point $p \in S$, pulling a massless thread connected to a marble with a force of magnitude T. The marble is constrained to move along the curve $C = \{\exp_p(\mathbf{x} + t\mathbf{w}) \mid t \in \mathbb{R}\}$ on the surface, like sliding on a frictionless wire shaped along that path. The marble has mass m, initial position $\exp_p(\mathbf{x})$, and initial velocity $(d\exp_p)_{\mathbf{x}}(\mathbf{w})$. Although the system is not in equilibrium, the thread has zero mass, so the net force applied to all portions of the thread must be zero. Thus, the thread remains a geodesic curve with constant tension T at every point, by the same reasoning as in Section 2. Therefore, the thread at time $t = 0$ can be modeled as

$$\exp_p(s\mathbf{x}), \quad 0 \le s \le 1. \tag{5}$$

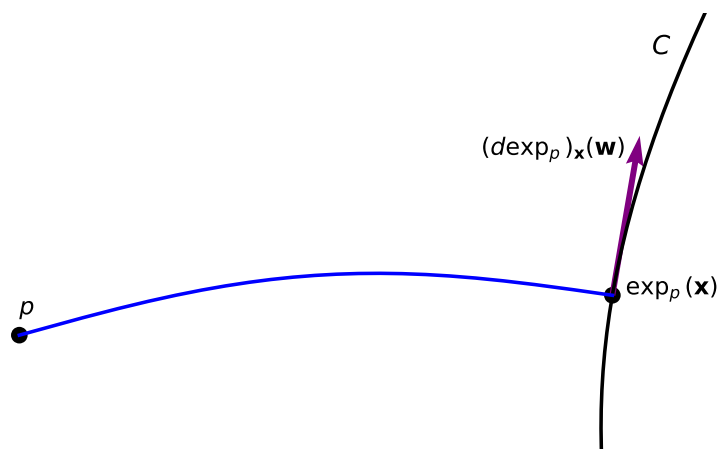

Figure 4. Initial state of the setting

After a short time Δt, the marble moves to

$$\exp_p(\mathbf{x}) + (d\exp_p)_{\mathbf{x}}(\mathbf{w})\,\Delta t + o(\Delta t). \tag{6}$$

By the definition of the differential

$$(d\exp_p)_{\mathbf{x}}(\mathbf{w}) = \frac{\exp_p(\mathbf{x} + \mathbf{w}\Delta t) - \exp_p(\mathbf{x})}{\Delta t} + o(1) \tag{7}$$

and the fact that the marble is constrained to lie on C, (6) is equal to

$$\exp_p(\mathbf{x} + \mathbf{w}[\Delta t + o(\Delta t)])(1). \tag{8}$$

Therefore, the new thread configuration can be modeled as

$$\exp_p(\mathbf{x} + \mathbf{w}[\Delta t + o(\Delta t)])(s), \quad 0 \le s \le 1. \tag{9}$$

Note that we are not claiming this configuration to be unique—geodesics between two points on a surface may not be—but rather that it is a viable scenario under Newtonian mechanics. This scenario is viable because the configuration in (9) is a geodesic, and the thread shapes (5) and (9) are close to each other when Δt is small. Therefore, this mechanical argument remains valid.

Now, we compute the change in kinetic energy:

$$\Delta K = (\mathbf{F} \cdot \mathbf{v})\, \Delta t + o(\Delta t) \tag{10}$$

$$= \left[-T \frac{(d\exp_p)_{\mathbf{x}}(\mathbf{x})}{\|(d\exp_p)_{\mathbf{x}}(\mathbf{x})\|} \right] \cdot (d\exp_p)_{\mathbf{x}}(\mathbf{w})\, \Delta t + o(\Delta t). \tag{11}$$

Since $\left\| \frac{d}{ds} \exp_p(s\mathbf{x}) \right\|$ is constant and equal to $\|\mathbf{x}\|$ for all s, the length of the thread in (5) is $\|\mathbf{x}\|$, while the length in (9) is $\|\mathbf{x} + \mathbf{w}\, \Delta t + o(\Delta t)\|$. Using this, the work done by the motor is

$$\Delta W = -T\, \Delta l \tag{12}$$

$$= -T\, (\|\mathbf{x} + \mathbf{w}\Delta t + o(\Delta t)\| - \|\mathbf{x}\|) \tag{13}$$

$$= -T\|\mathbf{x}\| \left[\left(1 + \frac{2\mathbf{x} \cdot \mathbf{w}\, \Delta t}{\|\mathbf{x}\|^2} + o(\Delta t) \right)^{1/2} - 1 \right] \tag{14}$$

$$= -\frac{T\, \mathbf{x} \cdot \mathbf{w}}{\|\mathbf{x}\|}\, \Delta t + o(\Delta t). \tag{15}$$

By the conservation of energy, we have $\Delta K = \Delta W$. Substituting $\|(d\exp_p)_{\mathbf{x}}(\mathbf{x})\| = \|\frac{d}{ds} \exp_p(s\mathbf{x})|_{s=1}\| = \|\frac{d}{ds} \exp_p(s\mathbf{x})|_{s=0}\| = \|\mathbf{x}\|$, dividing both sides by Δt, and taking the limit as $\Delta t \to 0$, we obtain:

$$(d\exp_p)_{\mathbf{x}}(\mathbf{x}) \cdot (d\exp_p)_{\mathbf{x}}(\mathbf{w}) = \mathbf{x} \cdot \mathbf{w}. \tag{16}$$

■

Jacobi's Theorem

Lemma 3. *Let $\boldsymbol{\gamma} : [0, L] \to S$ be a geodesic parametrized by arc length with $\boldsymbol{\gamma}(0) = p$ and $\boldsymbol{\gamma}(L) = q$. Let $\boldsymbol{\alpha} : [0, \tilde{L}] \to S$ be another arc-length parametrized curve with $\boldsymbol{\alpha}(0) = p$ and $\boldsymbol{\alpha}(\tilde{L}) = q$.*

Assume that for each $s \in (0, \tilde{L}]$ there exists a geodesic $\boldsymbol{\beta}_s : [0, L_s] \to S$, parametrized by arc length, such that $\boldsymbol{\beta}_s(0) = p$ and $\boldsymbol{\beta}_s(L_s) = \boldsymbol{\alpha}(s)$, and these geodesics depend smoothly on s. Assume further that $\boldsymbol{\beta}_{\tilde{L}} = \boldsymbol{\gamma}$. Then $L \le \tilde{L}$.

Proof. Place a motor at p pulling a massless thread with constant force T, with a marble attached to the free end. Since the thread is massless, its shape is geodesic. As the marble moves along $\boldsymbol{\alpha}$ from p to q, the thread takes the shape

$$\boldsymbol{\beta}_s(u), \quad 0 \le u \le L_s.$$

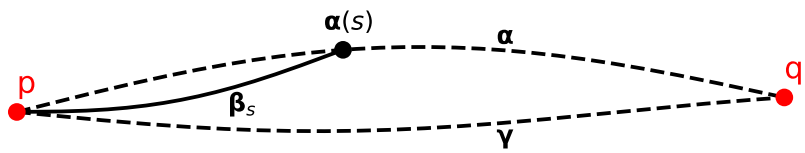

Figure 5. Jacobi's theorem

(Again, we emphasize this is just one physically admissible shape consistent with Newton's laws.)

The work done by the motor can be computed in two ways. First, by integrating the force along the marble's path:

$$W = \int \mathbf{F} \cdot d\mathbf{r} \tag{17}$$

$$= \int_0^{\tilde{L}} -T\,\boldsymbol{\beta}_s'(L_s) \cdot \boldsymbol{\alpha}'(s)\,ds \tag{18}$$

$$\geq -T \int_0^{\tilde{L}} ds = -T\tilde{L}, \tag{19}$$

where the inequality follows since $\boldsymbol{\beta}_s'(L_s)$ and $\boldsymbol{\alpha}'(s)$ are unit vectors.

Second, because the motor exerts constant force T and the thread length changes from its initial to final value L, the work is

$$W = -TL. \tag{20}$$

Comparing (19) and (20) yields $L \leq \tilde{L}$. ■

Theorem 4 (Jacobi's first theorem). *Suppose* $(d\exp_p)_{s\boldsymbol{\gamma}'(0)}(\mathbf{w}) \neq 0$ *for all* $s \in [0, L]$ *and all nonzero* $\mathbf{w} \in T_pS$. *Then there exists an open set* U *containing the image of* $\boldsymbol{\gamma}$ *such that every curve* $\boldsymbol{\alpha}$ *with endpoints* p *and* q *and image contained in* U *has length at least that of* $\boldsymbol{\gamma}$.

Proof. Since $(d\exp_p)_{s\boldsymbol{\gamma}'(0)}(\mathbf{w}) \neq 0$ for all $s \in [0, L]$ and all nonzero $\mathbf{w} \in T_pS$, standard arguments using compactness and the fact that $\exp_p$ is a local diffeomorphism show that $\exp_p^{-1}$ is defined on a neighborhood of the image of $\boldsymbol{\gamma}$.

For any curve $\boldsymbol{\alpha}$ whose image lies sufficiently close to that of $\boldsymbol{\gamma}$, and for each s, there exists a geodesic $\boldsymbol{\beta}_s$ with $\boldsymbol{\beta}_s(0) = p$ and $\boldsymbol{\beta}_s'(0) = \mathbf{x}_s/\|\mathbf{x}_s\|$, where $\mathbf{x}_s = (\exp_p)^{-1}(\boldsymbol{\alpha}(s))$. Each $\boldsymbol{\beta}_s$ satisfies the hypotheses of Lemma 3, so $\boldsymbol{\gamma}$ is locally length-minimizing between p and q; see [9, Section 5.9, Theorem 1] for details. ■

3. ISOPERIMETRIC INEQUALITY

Least Perimeter Curve has Constant Geodesic Curvature

Theorem 5. *Assuming the existence of a simple closed curve on a surface* S *that encloses a region of area* A *and minimizes the perimeter, such a curve must have constant geodesic curvature.*

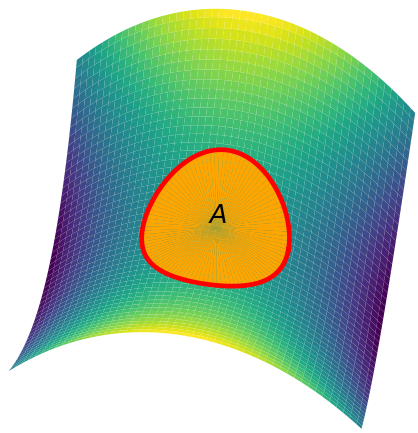

Figure 6. Isoperimetric Problem on Surface

Proof. Consider a massless motor that pulls a thread of nonzero mass with a force of magnitude T, where the end of the thread is attached to the motor itself. Suppose the thread lies on a surface S and forms a simple closed curve, enclosing an incompressible two-dimensional body of water of fixed area A.

Of course, a two-dimensional fluid does not exist physically, but it serves as a helpful idealization. For readers uncomfortable with this abstraction, one may alternatively replace the surface S with a narrow volume region $\{\, \mathbf{X} + t\mathbf{N}(\mathbf{X}) \mid \mathbf{X} \in S,\ 0 \le t \le \epsilon \,\}$ for some very small $\epsilon > 0$. In this interpretation, a thread of thickness ϵ bounds a three-dimensional incompressible fluid of volume approximately ϵA, distributed in a thin neighborhood of the surface.

The motor and the thread are constrained to remain on the surface S, but are free to move along it. The potential energy of the system is

$$U = T\,l, \tag{21}$$

where l is the perimeter of the curve. If the perimeter is minimized, then the potential energy is minimized, and the system is in equilibrium.

At equilibrium, the internal pressure p_w must be uniform throughout the enclosed region; otherwise, the fluid would spontaneously flow from regions of higher pressure to lower pressure, contradicting the assumption of equilibrium.

Let the curve be parametrized by arc length as $\mathbf{r} : [0, L] \to S$. Let $f(s)\mathbf{N}(s)\,\Delta s + o(\Delta s)$ represent the constraint force applied to the small thread segment from $\mathbf{r}(s)$ to $\mathbf{r}(s + \Delta s)$, as in the definitions given in Theorem 1, and let the unit normal vector $\mathbf{N}(s)$ be oriented such that $\mathbf{N}(s) \times \mathbf{r}'(s)$ points into the enclosed region.

Then, the net force acting on the small segment from $\mathbf{r}(s)$ to $\mathbf{r}(s + \Delta s)$ is

$$-T(s)\mathbf{r}'(s) + T(s + \Delta s)\mathbf{r}'(s + \Delta s) + f(s)\mathbf{N}(s)\,\Delta s - p_w\,\mathbf{N}(s) \times \mathbf{r}'(s)\,\Delta s + o(\Delta s) = 0. \tag{22}$$

Dividing by Δs and taking the limit as $\Delta s \to 0$ gives

$$T'(s)\mathbf{r}'(s) + T(s)\mathbf{r}''(s) + f(s)\mathbf{N}(s) - p_w\,\mathbf{N}(s) \times \mathbf{r}'(s) = 0. \tag{23}$$

Taking the inner product of both sides of (23) with $\mathbf{r}'(s)$ yields

$$T'(s) = 0, \tag{24}$$

so the tension is constant T along the curve. Taking the inner product of (23) with $\mathbf{N}(s) \times \mathbf{r}'(s)$ gives

$$\mathbf{r}''(s) \cdot (\mathbf{N}(s) \times \mathbf{r}'(s)) = \frac{p_w}{T}. \tag{25}$$

The left-hand side is the geodesic curvature $k_g(s)$ of the curve. Thus,

$$k_g(s) = \frac{p_w}{T}, \tag{26}$$

so $k_g(s)$ is constant.

■

Double Bubble Problem on a Surface

Theorem 6. *Let C_1, C_2, and C_3 be three curves on a surface S that share the same initial and terminal points, intersecting only at those two endpoints. Suppose C_2 lies between C_1 and C_3. Let the region enclosed by C_1 and C_2 have fixed area A_1, and the region enclosed by C_2 and C_3 have fixed area A_2. Among all such configurations, suppose the total perimeter—that is, the sum of the lengths of C_1, C_2, and C_3—is minimized. Then each of the three curves must have constant geodesic curvature, and the arcs must meet at angles of $120°$.*

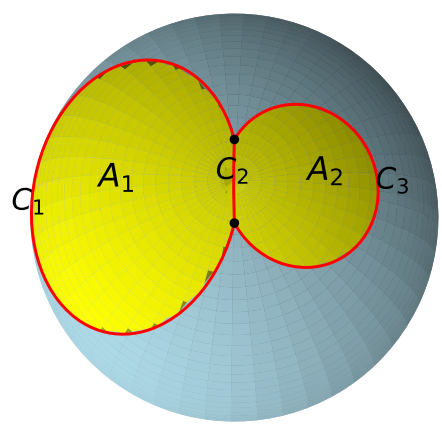

Figure 7. Double Bubble Problem

Proof. Consider a motor that pulls three threads, each attached to a single marble that moves freely but is constrained to lie on the surface S. Name the threads C_1, C_2, and C_3, and assume that the motor pulls each with the same constant tension T. Place incompressible two-dimensional water of area A_1 between C_1 and C_2, and water of area A_2 between C_2 and C_3. The potential energy of the system is proportional to the perimeter, given by $U = T(l_1 + l_2 + l_3)$, where l_i is the length of C_i. Thus, minimizing the perimeter corresponds to minimizing the potential energy, so the system must be in equilibrium.

By the same reasoning as in Theorem 5, each of C_1, C_2, and C_3 has constant geodesic curvature. At the junction point where the three arcs meet, force balance requires that the vector sum of the three tension forces (each of magnitude T) vanish, which occurs if and only if the three curves meet at angles of $120°$. ■

Additionally, we immediately observe that the geodesic curvature of C_2 equals the difference between the geodesic curvatures of C_1 and C_3 without resorting to tedious calculations.

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